Sanskriti Singh

B.Tech Information Technology Student

sans.vbps@gmail.com | +91 9515268515 | github.com/kritisans25 | linkedin.com/in/sanskriti-singh-2b0bb72b8 | leetcode.com/u/sans_0225

SUMMARY

Final-year Information Technology undergraduate (CGPA 8.7) building ML-driven full-stack applications and distributed data pipelines, with strong problem solving, analytical thinking, and attention to detail. Experienced with Python, Flask, React.js, Apache Spark, and Scala across projects processing millions of records, backed by IBM and Udemy certifications in DevOps, CI/CD, and cloud deployment. Comfortable with client communication, troubleshooting, and cross-functional/global collaboration in fast-paced, Agile settings. Seeking Software Development Engineer roles in backend engineering and applied machine learning.

TECHNICAL SKILLS

- **Languages:** Python, C, C++, Java, JavaScript
- **Core CS:** Data Structures & Algorithms, OOP, System Design, DBMS, Operating Systems, Computer Architecture
- **Web:** HTML, CSS, Bootstrap, React.js, Flask, REST APIs
- **Machine Learning:** Pandas, NumPy, Scikit-learn, XGBoost, OpenCV, OCR, Model Training & Evaluation
- **Big Data:** Apache Spark, Hadoop HDFS, MinHash LSH, Distributed Data Processing
- **DevOps & Cloud:** Docker, Kubernetes, Jenkins, CI/CD, GitHub Actions, Terraform, AWS
- **Databases:** SQL, Oracle SQL, SQLite
- **Tools & Practices:** Git, GitHub, Figma, Design Thinking, Microsoft Word, Excel, PowerPoint
- **Professional Skills:** Incident Response, Troubleshooting, Problem Solving, Analytical Thinking, Client Communication, Process Improvement, Attention to Detail, Global Collaboration

EDUCATION

Vellore Institute of Technology, Vellore — B.Tech in Information Technology **2023 – Present**
CGPA: 8.7

Kennedy High The Global School — Senior Secondary (XII), 94% 2022 – 2023
Vishwa Bharati School — Secondary (X), 98% 2020 – 2021

PROJECTS

AI-Driven Workplace Stress Monitoring System **2026**
Flask, Machine Learning, OpenAI API, SQLite, Chart.js

- Built an end-to-end web application to assess and monitor workplace stress using machine learning and physiological data.
- Developed a RandomForest-based stress prediction model achieving ~90% accuracy using questionnaire-driven inputs; integrated Google Fit API for real-time heart-rate anomaly detection with automated email alerts.
- Implemented an OpenAI-powered emotional support chatbot with mood detection and conversational memory, and designed interactive dashboards for stress trends and explainability insights.

CBSE Board Answer Sheet Evaluation System (Ongoing) **2026**
Machine Learning, Python, Flask, OpenCV, OCR

- Developing an AI-assisted system to automate evaluation of handwritten CBSE board exam answer sheets, including OCR/image preprocessing pipelines built with OpenCV.
- Implementing ML and NLP techniques to compare student answers against marking schemes, generate predicted scores, and troubleshoot scoring discrepancies for improved accuracy.
- Designing a web-based interface for uploads and results, with explainable scoring to highlight matched key points and improve process transparency.

AgriVision – Smart Crop Recommendation System **2024**
Machine Learning Project — Python, XGBoost, Flask

- Built an end-to-end ML pipeline to recommend crops and fertilizers based on soil nutrients and environmental parameters.
- Integrated real-time OpenWeatherMap data to improve prediction accuracy and exposed predictions via RESTful Flask APIs for frontend-backend communication.

CERTIFICATIONS

- **Decoding DevOps: From Basics to Advanced Projects** — Udemy, 2025 (63-hr program: Linux, Git, Docker, Kubernetes, Jenkins, GitHub Actions, Terraform, Ansible, AWS)
- **DevOps Fundamentals** — IBM Career Education Program, 2025
- **DevOps, Agile and Design Thinking** — IBM Career Education Program, 2025

LEADERSHIP AND ACTIVITIES

Design Head, Dream Merchants Club, VIT **Apr 2025 – Present**

- Lead design initiatives for technical and finance-focused events, collaborating with cross-functional teams to manage large-scale events such as Stochastic and Dream Team — from ideation and branding to execution.
- Coordinate deliverables in two-week Agile sprints with regular standups and review cycles across a multi-team organizing committee, emphasizing process improvement and clear communication.